

gdTAI v3: Round 12 vs Round 14

A checksum-pinned, same-cohort comparison across the 5.13-million-cell atlas, held-out atlas cohorts, and an independent external H5AD.

Selected: Round 14

Round 14 is selected because it is the only candidate that reaches the prespecified 0.80 full-atlas gold-recall floor while remaining below all FP guardrails. It also has substantially higher held-out cord-blood recall and higher mean F1 across the three evaluation frames. Round 12 remains the lower-FP fallback.

Round 12 atlas calls

213,182

Round 14 atlas calls

251,356

Shared calls

203,164

Prediction Jaccard

0.777

Executive Comparison

Round 14 adds 38,174 raw atlas calls and recovers 2,863 additional gold gdT cells. The cost is 2,832 additional paired-TCRAB calls and 2,201 additional NK-annotated calls. On the independent external set, Round 14 recovers 1 additional true gdT cell but adds 16 false positives, so Round 12 has the higher external F1. On the atlas held-out set, Round 14 has the substantially higher F1 because it rescues cord-blood gdT cells.

Measure	Round 12	Round 14	Round 14 minus Round 12
Raw atlas calls	213182	251356	38174
Gold gdT recovered	43681	46544	2863
Silver gdT recovered	1981	2031	50
Paired-TCRAB calls	2748	5580	2832
NK-annotated calls	1723	3924	2201

Evaluation	Round 12 precision	Round 12 recall	Round 12 F1	Round 14 precision	Round 14 recall	Round 14 F1
Full atlas primary gold	0.9823	0.7560	0.8544	0.9752	0.8056	0.8823
Atlas held-out combined	0.9925	0.6751	0.8036	0.9854	0.8530	0.9144
Independent external	0.9690	0.8576	0.9099	0.9509	0.8586	0.9024

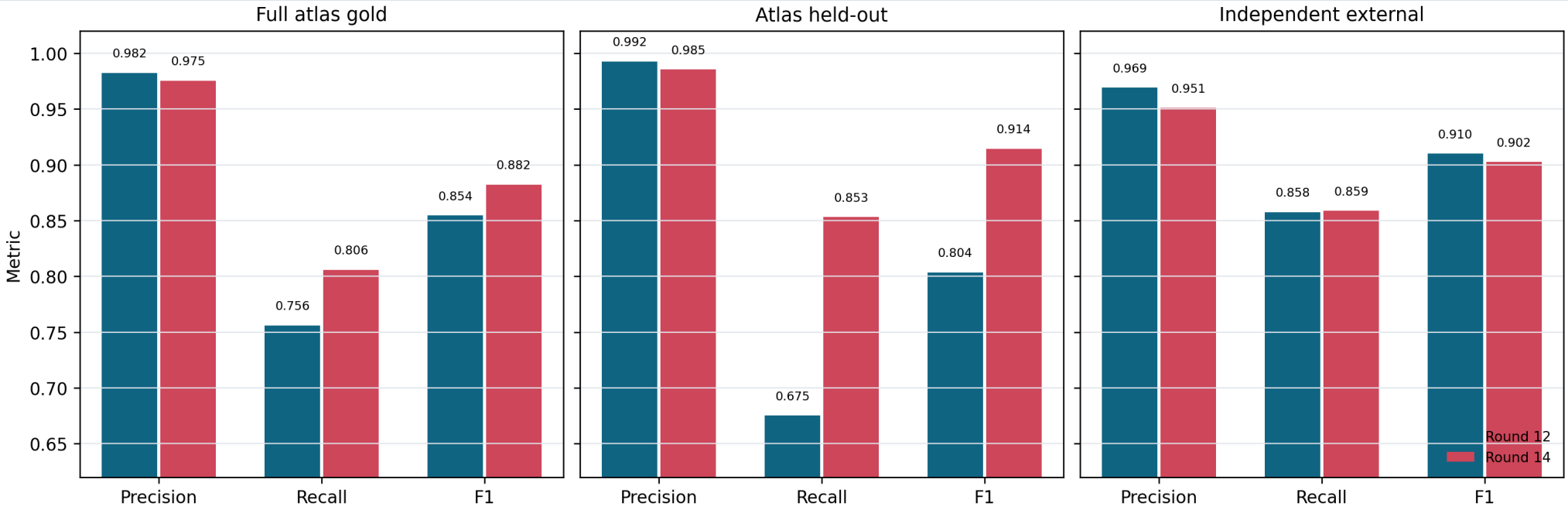
Decision Rule

Pass all four safety/recall guardrails, then maximize mean F1 across full-atlas gold, atlas-held-out, and independent-external evaluations.

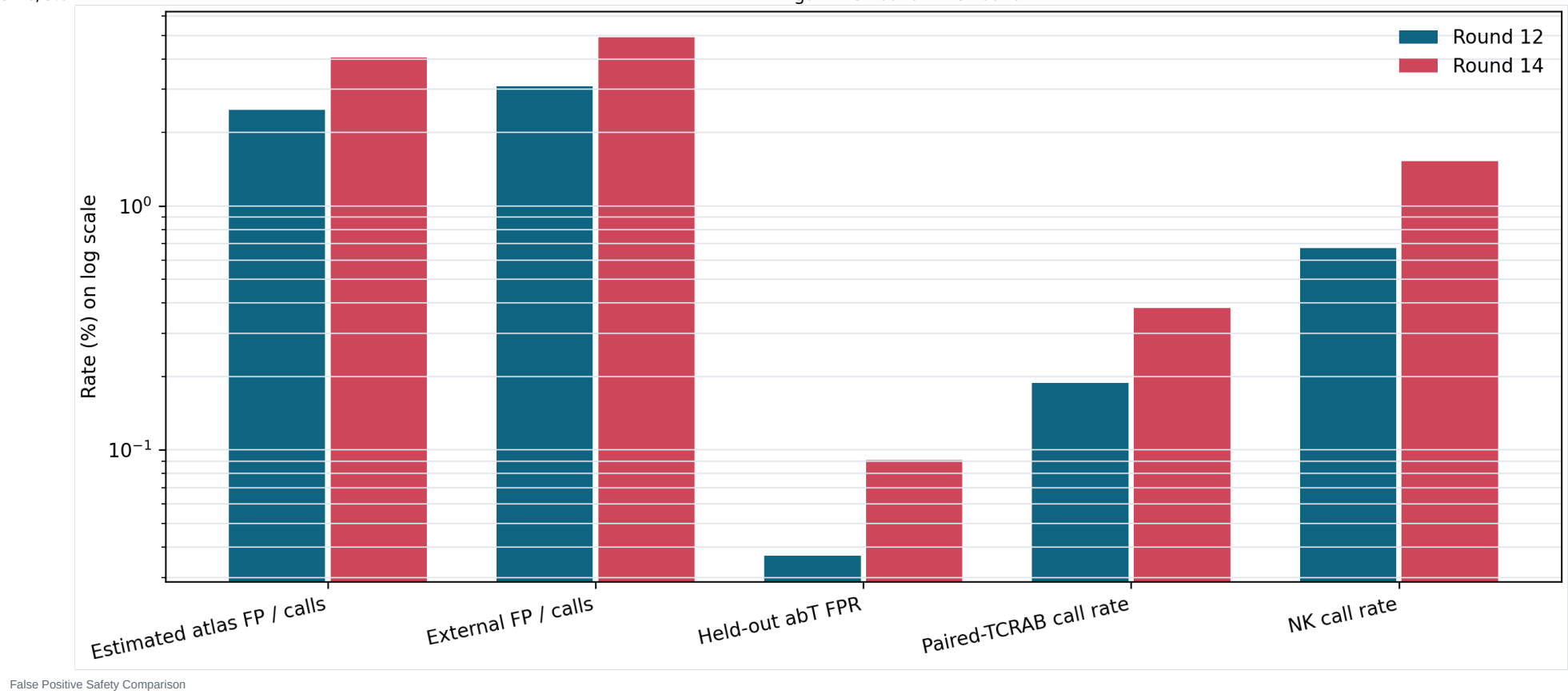
model	full_atlas_recall	full_atlas_estimated_fp_fraction	external_specificity	heldout_GSE254249_fp_rate	full_atlas_primary_gold_recall_ge_0p80	full_atlas_estimated_fp_fraction_le_0p05	external_specificity_ge_0p9985	heldout_GSE254249_fp_rate_le_0p001	all_guardrails_pass
round12	75.60%	2.47%	99.93%	0.04%	False	True	True	True	False
round14	80.56%	4.06%	99.89%	0.09%	True	True	True	True	True

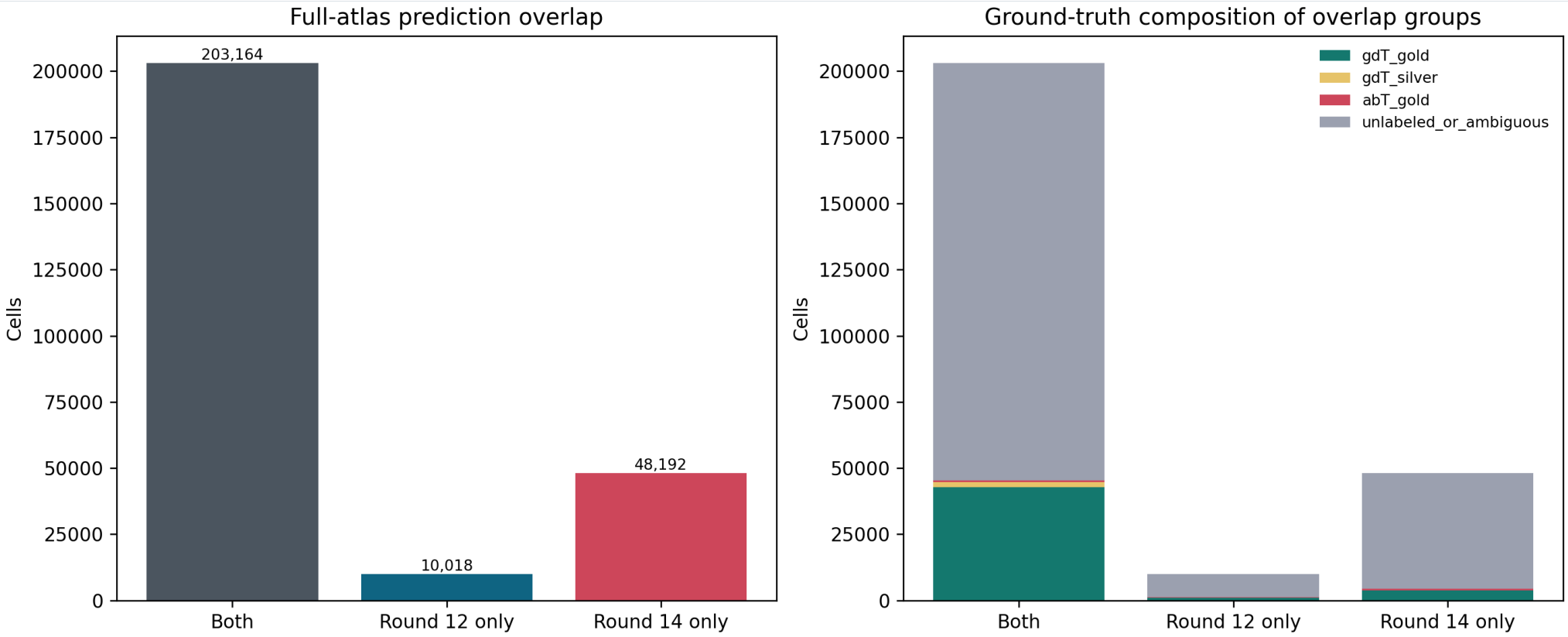
Mean F1 across full-atlas gold, atlas-held-out, and independent-external evaluations: Round 12 = 0.8560; Round 14 = 0.8997.

criterion	preferred_direction	round12	round14	winner
Full-atlas primary F1	higher	0.8544	0.8823	round14
Atlas held-out F1	higher	0.8036	0.9144	round14
Independent external F1	higher	0.9099	0.9024	round12
Full-atlas primary recall	higher	0.7560	0.8056	round14
Silver recall	higher	0.9029	0.9257	round14
Independent external precision	higher	0.9690	0.9509	round12
Full-atlas estimated FP fraction	lower	0.0247	0.0406	round12
Full-atlas paired-TCRAB calls	lower	2748.0000	5580.0000	round12
Full-atlas NK calls	lower	1723.0000	3924.0000	round12

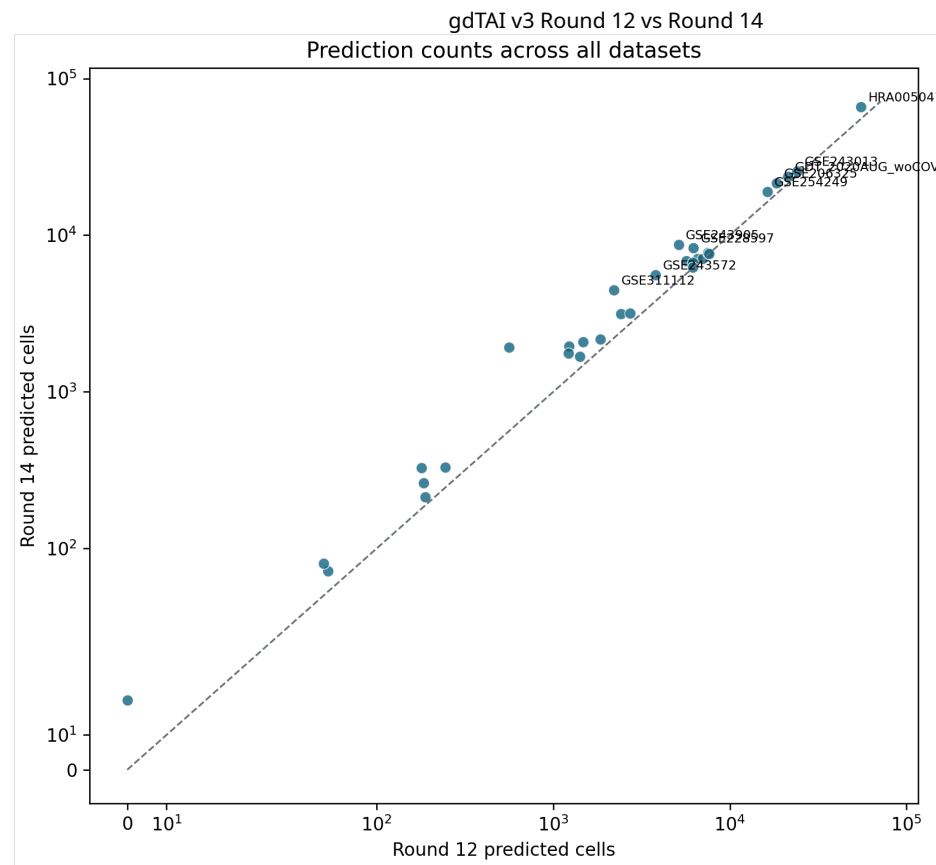


Performance Across Evaluations

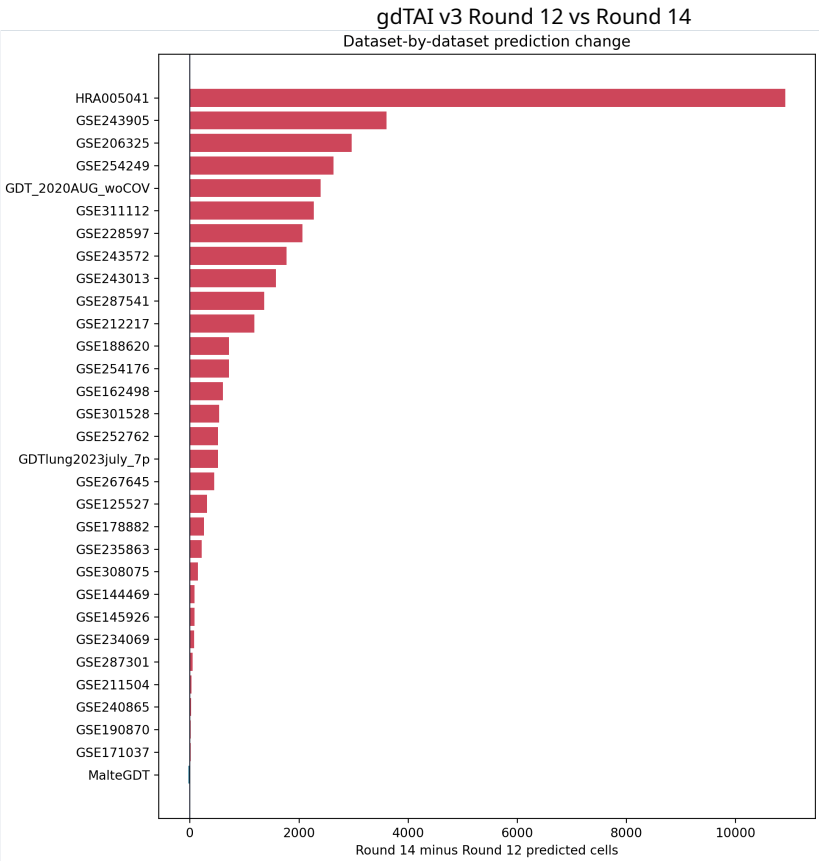




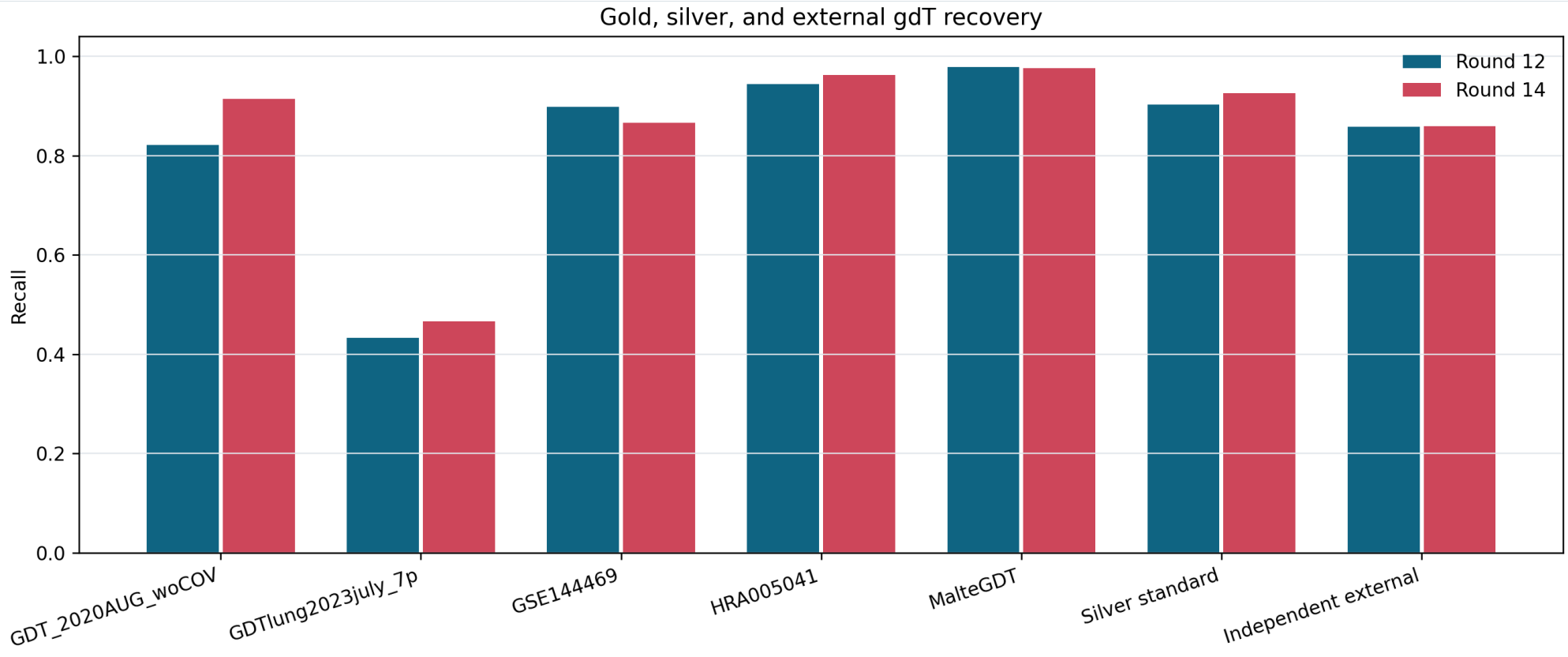
Prediction Overlap And Composition



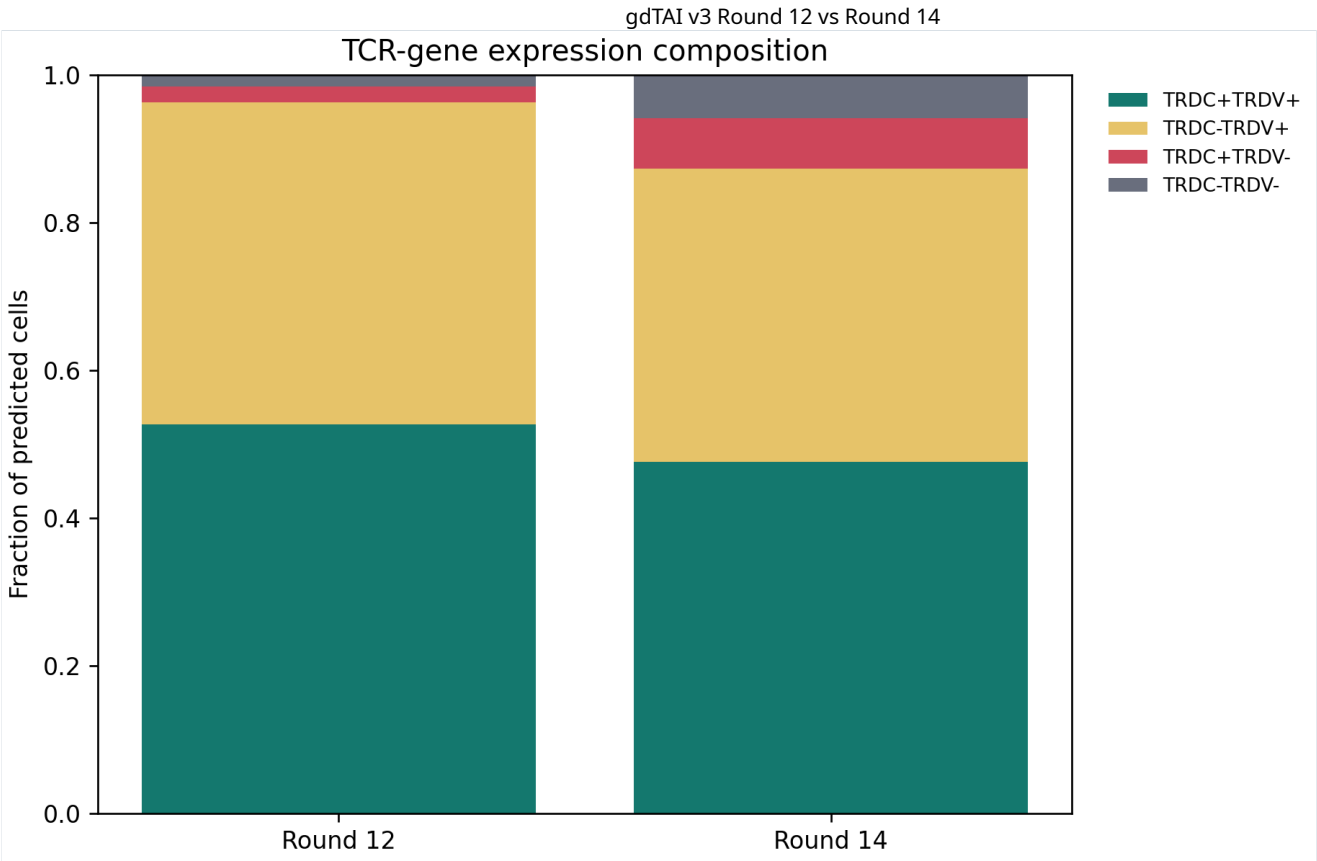
Per Dataset Prediction Scatter



Per Dataset Prediction Delta



Gdt Recall By Evidence Source



Predicted Tcr Quadrant Composition

Held-Out Cohorts

The atlas-held-out frame contains sorted cord-blood gdT positives from GDT_2020AUG_woCOV and paired-TCRAB/no-gdTCR negatives from GSE254249 . The external H5AD was not used for fitting or threshold selection.

strategy	group_value	n_cells	predicted_positive	tp	fp	fn	precision	recall	specificity	f1
round12	abT_GSE254249_paired_TCRAB_no_gdTCR	173697	64	0	64	0	0.0000	0.0000	0.9996	0.0000
round12	gdT_GDT_2020AUG_woCOV_cord_blood	12498	8437	8437	0	4061	1.0000	0.6751		0.8060
round14	abT_GSE254249_paired_TCRAB_no_gdTCR	173697	158	0	158	0	0.0000	0.0000	0.9991	0.0000
round14	gdT_GDT_2020AUG_woCOV_cord_blood	12498	10661	10661	0	1837	1.0000	0.8530		0.9207

model	group	group_type	n_cells	called_positive	rate
round12	all_real_gdT	positive_recall	948	813	0.8576
round12	TRDV_positive_real_gdT	positive_recall	847	793	0.9362
round12	TRDC_plus_TRDV_minus_real_gdT	positive_recall	48	17	0.3542
round12	Vd2_real_gdT	positive_recall	638	606	0.9498
round12	primary_negative	negative_false_positive	38047	26	0.0007
round12	NK_negative	negative_false_positive	920	3	0.0033
round12	paired_TCRAB_negative	negative_false_positive	33168	23	0.0007
round12	TRDC_plus_TRDV_minus_real_gdT	negative_false_positive	652	6	0.0092
round14	all_real_gdT	positive_recall	948	814	0.8586
round14	TRDV_positive_real_gdT	positive_recall	847	790	0.9327
round14	TRDC_plus_TRDV_minus_real_gdT	positive_recall	48	22	0.4583
round14	Vd2_real_gdT	positive_recall	638	600	0.9404

model	group	group_type	n_cells	called_positive	rate
round14	primary_negative	negative_false_positive	38047	42	0.0011
round14	NK_negative	negative_false_positive	920	9	0.0098
round14	paired_TCRAB_negative	negative_false_positive	33168	36	0.0011
round14	TRDC_plus_TRDV_minus_negative	negative_false_positive	652	14	0.0215

Prediction Overlap

total_atlas_cells	round12_predicted	round14_predicted	both	round12_only	round14_only	union	jaccard	fraction_round12_also_round14		fraction_round14_also_round12	
5128904	213182	251356	203164	10018	48192	261374	0.7773	0.9530		0.8083	

overlap_category	n_cells	unlabeled_or_ambiguous	unlabeled_or_ambiguous_fraction	abT_gold	abT_gold_fraction	gdT_gold	gdT_gold_fraction	gdT_silver	gdT_silver_fraction	NK_annotation	paired_TCRAB_non_gdT	TRDC+TRDV+	TRDC-TRDV+	TRDC+TRDV-	TRDC-TRDV-
Both	203164	157819	0.7768	636	0.0031	42752	0.2104	1957	0.0096	1334	2002	110470	87995	3848	851
Round 12 only	10018	8912	0.8896	153	0.0153	929	0.0927	24	0.0024	389	746	1809	4934	735	2540
Round 14 only	48192	43779	0.9084	547	0.0114	3792	0.0787	74	0.0015	2590	3578	9195	11928	13272	13797

Round 14-only calls contain 3,792 gold and 74 silver gdT cells. Round 12-only calls contain 929 gold and 24 silver gdT cells.

Dataset-by-Dataset Audit

All datasets are shown. Count changes are predictions, not biological abundance estimates; denominator-aware rates and truth recovery are included where labels exist.

source_gse_id	total_cells	round12_predicted	round14_predicted	round14_minus_round12	jaccard	gdT_gold	round12_gold_recall	round14_gold_recall	round12_NK	round14_NK	round12_paired_TCRAB	round14_paired_TCRAB
HRA005041	766639	55253	66167	10914	79.26%	4894	94.34%	96.16%	152	194	1109	3208
GSE243905	350438	5119	8726	3607	55.54%	0			191	532	147	510
GSE206325	501012	18458	21428	2970	78.37%	0			61	93	0	0
GSE254249	406494	16261	18895	2634	78.60%	0			138	54	64	158
GD T_2020AUG_woCOV	25904	21275	23677	2402	88.16%	25904	82.13%	91.40%	11	12	0	0
GSE311112	90599	2206	4479	2273	46.44%	0			15	19	9	10
GSE228597	149812	6222	8287	2065	70.25%	0			34	49	0	0
GSE243572	188217	3804	5576	1772	61.45%	0			424	1661	59	88
GSE243013	759436	24043	25620	1577	81.24%	0			127	154	0	0
GSE287541	110874	562	1926	1364	24.46%	0			9	25	0	0
GSE212217	213706	5673	6860	1187	75.66%	0			47	138	207	384
GSE188620	123218	1232	1954	722	56.79%	0			76	228	25	123
GSE254176	96953	2422	3142	720	73.33%	0			37	80	0	0
GSE162498	75880	1471	2081	610	63.91%	0			85	213	0	0
GSE301528	44105	1223	1764	541	66.78%	0			55	124	0	0
GSE252762	45132	6187	6708	521	85.25%	0			2	4	0	0
GD Tlung2023july_7p	15175	6561	7081	520	80.74%	15175	43.24%	46.66%	17	25	0	0
GSE267645	92501	2730	3182	452	77.64%	0			32	79	0	0
GSE125527	46396	1848	2168	320	79.69%	0			3	23	0	0
GSE178882	67755	1412	1675	263	75.00%	0			24	89	0	0
GSE235863	241766	7506	7724	218	77.96%	0			15	24	0	0
GSE308075	112337	180	328	148	47.25%	0			0	2	4	32
GSE144469	107068	7018	7103	85	83.34%	4003	89.78%	86.64%	70	5	1124	1067
GSE145926	21039	245	330	85	52.93%	0			43	23	0	0
GSE234069	20807	184	262	78	63.37%	0			1	2	0	0
GSE287301	340227	6161	6216	55	77.50%	0			27	32	0	0
GSE211504	37244	50	80	30	46.07%	0			0	0	0	0
GSE240865	21680	189	213	24	77.09%	0			10	16	0	0
MalteGDT	7800	7634	7612	-22	98.57%	7800	97.87%	97.59%	3	3	0	0
GSE190870	16753	0	20	20	0.00%	0			0	20	0	0
GSE171037	31937	53	72	19	32.98%	0			14	1	0	0

Paired Statistical Tests

Exact McNemar tests compare discordant cell-level errors. P-values are descriptive because cells from the same donor are correlated; effect sizes and cohort consistency remain the primary evidence.

evaluation	n_cells	round12_errors	round14_errors	round12_correct_round14_wrong	round12_wrong_round14_correct	discordant	exact_mcnemar_p	avored_model	note
independent_external_primary	38995	161	176	47	32	79	0.1147	round12	Cell-level paired test; descriptive because cells within donors are correlated.
independent_external_gdT	948	135	134	21	22	43	1.0000	round14	Cell-level paired test; descriptive because cells within donors are correlated.
independent_external_negative	38047	26	42	26	10	36	0.0113	round12	Cell-level paired test; descriptive because cells within donors are correlated.
atlas_heldout_combined	186195	4125	1995	286	2416	2702	0.0000	round14	Cell-level paired test; descriptive because cells within donors are correlated.
atlas_heldout_abT_GSE254249_paired_TCRAB_no_gdT	173697	64	158	127	33	160	0.0000	round12	Cell-level paired test; descriptive because cells within donors are correlated.
atlas_heldout_gdT_GDT_2020AUG_woCOV_cord_blood	12498	4061	1837	159	2383	2542	0.0000	round14	Cell-level paired test; descriptive because cells within donors are correlated.
full_atlas_gdT_gold	57776	14095	11232	929	3792	4721	0.0000	round14	Cell-level paired test; descriptive because cells within donors are correlated.
full_atlas_gdT_silver	2194	213	163	24	74	98	0.0000	round14	Cell-level paired test; descriptive because cells within donors are correlated.
full_atlas_abT_gold	603203	789	1183	547	153	700	0.0000	round12	Cell-level paired test; descriptive because cells within donors are correlated.

Artifact and Cache Verification

Both artifacts were frozen before comparison. Direct external inference was required to match the existing validated cache exactly. A deterministic atlas sample included shared calls, each model-only group, and cells called by neither model.

model	display_name	payload_model	threshold	sha256	size_bytes	snapshot_path
round12	Round 12	v3_round12_hist_gradient_fixed_Op5	0.5000	7373e79350f7db190c415b376b9763e31652754438ee8c5afd3853beb7b2ebc4	808096	/home/tanlikai/databank/publicdata/tools/output_geo_tcell_research/integrated_dataset/models/gdT_prediction_classifier/gdtai_v3_round12_vs_round14/round12_model.pkl
round14	Round 14	v3_round14_v2_score_trdc_gate_fixed_fp936	0.9360	16dedc0081da9b8487887341232bc8c9c9403dd3bbd72e04cab43d4cd7b2e09	15532	/home/tanlikai/databank/publicdata/tools/output_geo_tcell_research/integrated_dataset/models/gdT_prediction_classifier/gdtai_v3_round12_vs_round14/round14_model.pkl

check	model	n_cells	score_match	prediction_match	n_sampled	n_matching	n_mismatching
external_direct_vs_cache	round12	46273.0000	True	True			
external_direct_vs_cache	round14	46273.0000	True	True			
atlas_direct_artifact_vs_cell_cache	round12		NaN	True	2000.0000	2000.0000	0.0000
atlas_direct_artifact_vs_cell_cache	round14		NaN	True	2000.0000	2000.0000	0.0000

Interpretation

Round 14 should remain the canonical balanced model. Its recall gain is large in the fully held-out cord-blood cohort and sufficient to clear the established 0.80 atlas gold-recall floor. Its independent external recall is essentially unchanged relative to Round 12, while external precision is lower. This means Round 14 is not uniformly superior: Round 12 is the appropriate documented high-purity fallback when false-positive minimization is more important than sorted-gdT recovery.

The Round 14 increase in NK and TRDC+TRDV- calls remains a real caveat. Downstream atlas construction should continue removing cells with direct paired-TCRAB/no-gdT evidence and should report NK-annotation sensitivity analyses. The comparison does not justify treating every unlabeled Round 14-only cell as a confirmed gdT cell.

Reproducibility

- Atlas H5AD: `/home/tanlikai/databank/publicdata/tools/output_geo_tcell_research/high_speed_temp/Integrated_dataset/integrated_plus6.h5ad`
- External H5AD: `/home/tanlikai/databank/owndata/singlecell/data/results/phase4_final_annotated.h5ad`
- No H5AD was modified.
- Detailed CSV tables: `/home/tanlikai/databank/publicdata/tools/output_geo_tcell_research/Integrated_dataset/tables/gdT_prediction/gdtai_v3_round12_vs_round14`
- Model snapshots: `/home/tanlikai/databank/publicdata/tools/output_geo_tcell_research/Integrated_dataset/models/gdT_prediction_classifier/gdtai_v3_round12_vs_round14`

Generated from `workflows/gdtai/compare_gdtai_v3_round12_vs_round14.py`.